

2(a)	<i>max 1 in molecule section if mps from collagen fibre included as features of a molecule</i> <i>max 1 in fibre section if mps from molecule included as features of collagen fibre</i> or <i>if polypeptide described in molecule section and all features in fibre section are descriptive for a collagen molecule</i> <i>any five from:</i> <i>molecule (to max 4)</i> 1 three, helical / AW, polypeptides / chains I quaternary structure or triple helix ; 2 (helical chains) tightly wound ; 3 hydrogen bonds between, polypeptides / chains ; 4 (generally) every third amino acid (in each, polypeptide / chain) is glycine ; <i>fibre (to max 4)</i> 5 cross links / covalent bonds , between molecules ; 6 molecules arranged in parallel ; 7 molecules (in fibril / fibre) are, staggered / ends not aligned / AW ; 8 AVP ; e.g. detail of H bonds collagen fibre can be composed of collagen fibrils made of (parallel) collagen molecules glycosylated (amino acids in polypeptide) / hydroxylated	5
2(b)(i)	α -glucose / alpha glucose ;	1
2(b)(ii)	label to glycosidic bond labelled G ;	1
2(b)(iii)	NH ₂ CH ₂ CHCH ₂ CH ₂ circled <u>and</u> labelled R ;	1
2(c)	correct named tissue or structure ; e.g. blood vessels / cartilage / bone / ligaments / skin / tendons (is/are) weaker / deformed / easily broken / AW ; A description of how function is affected	2